

SOC Incident Response Report

1. Executive Summary

A simulated cyber attack involving a Samba exploit was conducted using Metasploit and Caldera to evaluate the organization's detection and response capabilities. The attack originated from IP address 192.168.56.3 targeting a vulnerable Metasploitable system.

Wazuh successfully detected multiple alerts including unauthorized access attempts and privilege escalation. The incident was analyzed and contained by blocking the attacker's IP using iptables and CrowdSec.

The overall response demonstrated effective detection; however, improvements are required in response time and security controls to prevent similar attacks in the future.

2. Incident Timeline

Timestamp	Event
16:09:09	Samba exploit attempt detected
16:09:21	Privilege escalation to root
16:09:24	Unauthorized access detected
16:10:00	Alert generated in Wazuh
16:12:00	IP blocked using iptables
16:13:00	System isolated

3. Detection & Analysis

Wazuh SIEM detected the attack through custom rules configured in local_rules.xml. Alerts included:

- Samba exploit attempt
- Unauthorized access
- Privilege escalation to root

The alerts were mapped to MITRE ATT&CK technique:

- T1210 – Exploitation of Remote Services

Elastic dashboards were used to visualize metrics such as MTTD and MTTR.

4. Response & Containment

The incident response actions included:

- Blocking attacker IP (192.168.56.3) using iptables
- Isolating the compromised system
- Creating a case in TheHive for tracking

Verification was performed using ping tests, confirming successful containment.

5. Root Cause Analysis (RCA)

The root cause of the incident was identified using the Fishbone Diagram and 5 Whys method.

Key causes include:

- Misconfigured Samba service
- Lack of patch management
- Weak security monitoring initially
- Poor user awareness

6. Metrics

- MTTD (Mean Time to Detect): 2 hours
- MTTR (Mean Time to Respond): 4 hours
- False Positive Rate: 20%

These metrics indicate good detection capability but moderate response delay.

7. Recommendations

- Apply regular patch updates to services like Samba
- Implement stricter firewall and access controls
- Improve detection rules in Wazuh to reduce false positives
- Conduct security awareness training
- Automate response using SOAR tools like TheHive

8. Conclusion

The simulation demonstrated a complete SOC workflow including detection, analysis, response, and reporting. While detection was effective, improvements in response automation and preventive controls are necessary to strengthen overall security posture.